Fanze Meng

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github.com/DMPoolM | LinkedIn

Computational materials MSc student with training in materials science and engineering mathematics. Experience in materials property prediction, uncertainty estimation, research software, and generative AI for industrial engineering.

EDUCATION

Institute of Science Tokyo Oct 2026 - present

MSc | Chemical Science / Computational Materials

Chalmers University of Technology Aug 2023 - Jun 2026

MSc | Engineering Mathematics and Computational Science

Queen Mary University of London Aug 2019 - Jul 2023

BEng | Materials Science and Engineering

SELECTED RESEARCH AND ENGINEERING EXPERIENCE

Volvo Cars | Materials & AI internship Feb 2025 - Jan 2026

- Built CNN-Transformer models to predict stress-strain curves from microstructure images of 90 representative volume elements, reaching validation R^2 = 0.89; estimated uncertainty with Monte Carlo dropout.
- Developed Material Centre, a Streamlit and Plotly application for materials similarity search using PCA and distance metrics; containerised with Docker and deployed for internal user testing.
- Developed a physics-informed model variant with elastic-modulus, smoothness, and boundary-condition losses.

Tohoku University | COLABS research exchange Sep 2024 - Feb 2025

- Built a graph-similarity clustering pipeline for 3D electron diffraction structure candidates, reducing the number requiring manual verification by 97.2% on average across 7 datasets (2,127 structures).
- Replicated a catalyst-screening study using pretrained FAIR-Chem GNN models; generated adsorbate-slab configurations, ran ML relaxations, and compared adsorption energies with literature values.

Institute of Science Tokyo | MSc research Oct 2026 - present

- Developing physics-aware generative models to reconstruct 3D atomic structures from pair distribution functions, targeting solid electrolytes for batteries.

Chalmers × Volvo Cars | MSc thesis Oct 2025 - Jun 2026

- Co-developed an LLM agent with retrieval over ABB manuals to generate RAPID robot code; implemented an HPC-based validation framework and benchmarked on approximately 50 manufacturing tasks.

TECHNICAL SKILLS AND LANGUAGES

ML and software: Python, PyTorch, TensorFlow, scikit-learn, XGBoost; CNNs, GNNs, PINNs; uncertainty estimation, LLM agents, RAG; Git, Docker, Azure, Streamlit, Plotly.

Materials: FAIR-Chem, ASE, RDKit; XRD, IR spectroscopy, SEM, electrochemical testing.

Languages: Chinese (native); English (fluent, IELTS 7.5); Japanese (actively studying).

SELECTED PUBLICATIONS

Second author: *Nano Today* (2023), doi: 10.1016/j.nantod.2023.101766.

Second author: *Journal of Physics: Conference Series* (2021), doi: 10.1088/1742-6596/2021/1/012021.